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**Comparative Efficacy of Rod Mills versus Ball Mills
in Grinding Kinetics: A Particle Size Distribution Analysis**

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Executive Summary

This report presents a comparative investigation into the grinding performance of three milling media — steel rods, steel balls, and ceramic balls — using silica sand as the feed material. The investigation was conducted as part of the IVP 360S practical module and is framed within the theoretical framework of Coulson and Richardson's Chemical Engineering, Volume 2 (5th Edition).

Particle size distribution (PSD) analysis was performed before and after a 10-minute milling cycle, with sieve analysis used as the primary measurement method. Energy requirements for each grinding medium were estimated using Rittinger's Law, with the characteristic particle sizes (L_1 and L_2) derived from the sieve fractions with the highest mass concentration.

Results indicated that rod milling produced the most effective size reduction, achieving a characteristic product size of approximately 400 μm compared to 600 μm for both steel and ceramic ball mills. The rod mill's superior performance is attributed to the screening effect — a geometric phenomenon in which the parallel arrangement of rods preferentially crushes larger particles while allowing finer particles to pass through, resulting in a more controlled product size distribution.

Ceramic balls proved the least effective grinding medium, a result attributed primarily to their lower density and hardness relative to steel, which reduces the impact energy transferred per collision. Steel balls demonstrated intermediate performance, confirming that media density and hardness are critical parameters in grinding efficacy.

Rittinger's Law calculations indicate that the rod mill required approximately 2.25 times the specific energy input of the ball mills to achieve grinding — a result consistent with its finer product size. It is recommended that rod mills be employed where a controlled, uniform product size is required, and that further investigation consider longer milling durations and alternative feed materials to validate the observed trends.

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Abbreviations and Glossary

PSD	— Particle Size Distribution
C&R	— Coulson & Richardson's Chemical Engineering
L₁	— Characteristic feed particle size (μm)
L₂	— Characteristic product particle size (μm)
E	— Specific energy requirement (J/kg)
K_r	— Rittinger's constant (0.0179 m ² /J)
F_c	— Crushing strength of material (30 MPa for silica)
μm	— Micrometre (1 × 10 ⁻⁶ m)

1. Introduction: Background and Motivation

Comminution — the systematic reduction of solid particle size — is a foundational unit operation in chemical engineering. Industrial raw materials are rarely encountered at the particle sizes required for downstream processing; large rocks and coarse aggregates must be reduced through a sequence of crushing and grinding stages before they are suitable for use. Comminution is therefore embedded in almost every manufacturing and mineral processing pipeline, from cement production to pharmaceutical manufacturing. (Coulson, 2002) page 95

Beyond preparing materials for processing, comminution serves several critical functions: it increases specific surface area to accelerate chemical reactions, ensures uniform particle size distributions for improved separation efficiency, and reduces particle dimensions to levels compatible with unit operations such as fluidised bed reactors, where excessive particle size can impede gas flow and compromise process performance. (Coulson, 2002)

This investigation focuses specifically on fine grinding — the size reduction of particles already in the millimetre to sub-millimetre range — and compares three grinding media: steel rods, steel balls of uniform size, and ceramic balls of mixed sizes. Silica sand (SiO_2) was selected as the feed material due to its availability, relatively uniform chemical composition, and widespread industrial relevance in sectors such as glass manufacturing and cement production (Corning, 2025). Silica has a hardness of 7 on the Mohs scale, making it a demanding but realistic test material for evaluating grinding performance. (Vaishnavi, n.d.)

The theoretical framework underpinning this investigation is drawn from Coulson and Richardson's Chemical Engineering, Volume 2 (5th Edition), specifically the treatment of particle technology, comminution mechanics, and energy laws governing size reduction. The primary analytical lens is particle size distribution (PSD) analysis, supplemented by energy requirement estimates derived from Rittinger's Law.

1.1 Objectives

The objectives of this investigation were as follows:

- To investigate the screening effect of rod mills in producing a more uniform product with fewer fines compared to ball mills.

- To determine the comparative efficacy of a uniform ball charge (steel balls, approximately equal sizes) against a uniform rod charge for the selected feed material.
- To determine the comparative efficacy of a graded ball charge (ceramic balls, mixed sizes) against a uniform rod charge for the selected feed material.
- To evaluate the specific energy consumption per unit of new surface area created for each grinding medium, using Rittinger's Law as the estimation framework.

2. Literature Review

The following section links the theoretical framework to the experimental investigation. It draws primarily from Coulson and Richardson's Chemical Engineering, Volume 2 (5th Edition) and serves to contextualise the equipment, materials, and analytical methods used.

2.1 Purpose and Industrial Context of Size Reduction

In large-scale industrial operations, raw materials are seldom encountered in the particle sizes required for a given process. It is common for materials to arrive as large rocks or coarse lumps, requiring progressive reduction through crushing and grinding stages before they can be processed further. The machinery selected for each stage is determined by the target particle size range and the physical properties of the material, particularly its hardness and fracture behaviour (Coulson & Richardson, Vol. 2). (Dzoneding mining and construction, 2025)

Size reduction enables several downstream advantages: it increases specific surface area, which directly accelerates the rate of heterogeneous chemical reactions; it improves the homogeneity of particle size distributions, which is important for consistent product quality; and it allows more effective separation of mixed-species materials, since differences in density, charge, and other physical properties become more pronounced at smaller particle (Vaishnavi, n.d.) (Coulson, 2002)

In fluidised bed reactors, particle size control is particularly critical — particles that are too large impede the free flow of gas through the bed, reducing mass transfer efficiency and potentially disrupting the hydrodynamic regime entirely. The preparation of fertilisers, pigments, and ceramics similarly depends on controlled particle size distributions achievable only through deliberate comminution (Coulson, 2002) page 95-96

2.2 Comminution Mechanics

According to (Coulson, 2002) page 98-99, four fundamental breakage mechanisms operate in comminution equipment: impact, compression, shear, and attrition. In practice, most industrial mills employ a combination of these mechanisms simultaneously, and it is rarely possible to isolate the dominant mode for a given machine. For ball and rod mills, the primary mechanisms are impact (from falling media) and attrition (from media rolling against the mill shell and each other).

An important distinction exists between the fracture behaviour of larger and smaller particles. For coarser particles, breakage is largely governed by the geometry and mechanical energy of the crushing event, and product size can be directly correlated with machine type and operating conditions. For finer particles, however, fracture behaviour is increasingly influenced by the internal structure of the material — specifically the distribution of grain boundaries, micro-cracks, and crystallographic planes along which fracture is energetically favoured (Coulson & Richardson, Vol. 2).

Silica (quartz) is a particularly relevant material in this context. Quartz exhibits a strong tendency toward agglomeration at fine particle sizes due to electrostatic surface forces, which can reduce the efficiency of ball milling but is less problematic in rod mills, where the open geometry between parallel rods limits aggregate buildup. This is consistent with the literature noting that rod mills are generally preferred for materials that tend to aggregate during grinding. (Vaishnavi, n.d.)

2.3 Ball Mills

Ball mills are classified as fine crushers and consist of a rotating hollow cylinder with an abrasion-resistant lining, which in some designs includes internal lifter bars that enhance the lifting and cascading action of the charge. In industrial applications, ball mills are used to grind feed material typically up to 50 mm in size. The grinding media — spherical balls — generally occupy 30 to 50 percent of the mill volume, with ball diameters ranging from 12 to 125 mm. (Coulson, 2002)

During operation, balls undergo progressive wear and are periodically replaced, resulting in a mill that contains balls across a range of sizes and ages at any given time. This is operationally beneficial: larger balls provide the impact energy required to break coarser particles, while smaller

balls fill the interstitial spaces and are responsible for producing finer product fractions. In smaller laboratory-scale mills, pebbles are sometimes substituted for steel balls (Coulson, 2002).

2.4 Rod Mills

Rod mills employ steel rods — typically approximately 50 mm in diameter — as the grinding medium. The rods extend the full length of the mill and lie parallel to the mill axis. Feed size generally does not exceed 25 mm, and rod mills are not well-suited to tough or highly abrasive materials. Their primary advantage is in grinding materials that tend to agglomerate, where the open geometry between parallel rods allows fines to escape without being over-ground (Coulson, 2002) (王, 2025).

2.5 The Screening Effect in Rod Mills

The screening effect is a phenomenon arising from the geometry of the rod charge. As the rods lie in parallel alignment, they create slots of consistent width between adjacent rods. Larger particles are selectively trapped in these slots and subjected to crushing, while particles already smaller than the slot width pass through largely unaffected. This selective grinding mechanism produces a narrower product size distribution compared to ball milling, with fewer ultra-fine particles (fines) in the product. The screening effect is a key reason why rod mills are preferred when a uniform coarse-to-medium product is required (Coulson & Richardson, Vol. 2). (王, 2025)

2.6 Energy Laws for Size Reduction

The energy required for comminution cannot be determined with high precision, but empirical laws have been developed to provide useful estimates. The two earliest formulations are Kick's Law and von Rittinger's Law, both derived from the same differential equation relating the incremental energy requirement to particle size (Coulson & Richardson, Vol. 2). The governing differential equation is:

$$dE/dL = -C / L^n \quad (\text{Eq. 2.1}) \quad (\text{Coulson, 2002})$$

Figure 1 Energy equation derivation

where E is the specific energy (J/kg), L is the characteristic particle size (m), C is a material-dependent constant, and n is an exponent that determines the law. For Rittinger's Law, $n = 2$, which gives:

$$E = K_r \cdot F_c \cdot (1/L_2 - 1/L_1) \quad (\text{Eq. 2.2}) \quad (\text{Coulson, 2002})$$

Figure 2 Energy Equation derivation 2

where K_r is Rittinger's constant ($0.0179 \text{ m}^2/\text{J}$ (Food Tech Notes, 2020)), F_c is the crushing strength of the material (30 MPa for silica (K. M. A. S. Bandara, 2022)), L_1 is the characteristic feed size, and L_2 is the characteristic product size. Rittinger's Law is most appropriate for fine grinding, where the energy input is primarily associated with the creation of new surface area rather than plastic deformation or fracture of large particles (Coulson, 2002) page 100-101. This is consistent with the particle size range encountered in this investigation (400–1000 μm).

3. Research Methodology

The experimental investigation was conducted at laboratory scale, following a standardised procedure to ensure consistency across the three grinding media. All measurements were performed using the same equipment under equivalent conditions.

3.1 Equipment

The key equipment used in this investigation was:

- Laboratory-scale rotating mill
- Steel grinding rods
- Steel grinding balls (uniform size)
- Ceramic grinding balls (mixed sizes)
- Sieve stack with shaker
- Electronic balance
- Sample collection containers

3.2 Variables

Independent variable: Grinding medium type (steel rods, steel balls, ceramic balls).

Dependent variable: Particle size distribution and mass retained per sieve fraction after grinding.

Control variables: Milling duration (10 minutes), initial sample mass (500 g), and mill rotational speed (assumed constant for all runs).

3.3 Procedure

The following procedure was followed for each grinding medium:

1. Sieve tare masses were recorded for each sieve in the stack.
2. The sample container was weighed, and the electronic balance was zeroed (tared).
3. 500 g of silica sand was measured into the container.
4. The sand was transferred to the sieve stack and shaken for 2 minutes to obtain the pre-milling particle size distribution.
5. The mass retained on each sieve was recorded individually.
6. The sand was collected into the container and its mass was recorded.
7. The mill was opened and the sand was placed inside together with the designated grinding medium.
8. The mill was operated for 10 minutes.
9. The mill was opened and the grinding medium and sand were collected into the container.
10. The sand was carefully brushed off the grinding medium and collected.
11. The sand was placed into the sieve stack and shaken for 2 minutes.
12. The mass retained on each sieve was recorded individually and the total collected mass was recorded.

3.4 Data Collection Notes

It is important to note that the particle size distribution data for the rod mill and steel ball mill runs were obtained from concurrent experimental groups within the same practical session, all following the identical procedure described above. The ceramic ball mill data was collected directly by this group. Cross-group data were used in accordance with the practical session design, which was intended to maximise the range of experimental conditions covered within the available time.

Minor losses of sample material were observed during transfer between stations, primarily in the form of fine airborne particles generated upon opening the mill. This represents a potential source of systematic error and is discussed further in Section 4.4.

3.5 Safety

Throughout the practical, appropriate personal protective equipment (PPE) was worn at all times. Dust masks were used to prevent inhalation of fine silica particles, which pose a respiratory health risk upon prolonged exposure. Nitrile gloves were worn to minimise skin contact with the feed and product material. Laboratory coats, closed shoes, and long trousers were worn to minimise skin exposure. All equipment was handled in accordance with the laboratory safety protocols.

4. Data Analysis and Calculations

This section presents the particle size distribution data, the energy requirement calculations based on Rittinger's Law, and a discussion of the observed trends and sources of error.

4.1 Particle Size Distribution Data

Table 4.1 — Pre-Milling Sieve Analysis Results (Initial Feed, 500 g) for ceramic (different sizes) and steel balls

Sieve Size (µm)	Steel balls (g)	Ceramic balls (g)
1180	844,82	812
600	347,14	378
425	300,81	316
250	300,8	291
150	292,62	292
75	256,99	263
Pan	451,72	259

(Repeat table format for Table 4.2, 4.3, 4.4 with post-milling data for each medium)

4.2 Energy Requirement Calculations (Rittinger's Law)

Rittinger's Law was applied to estimate the specific energy requirement for each grinding medium. The characteristic feed size L_1 was taken as 1000 µm (1 mm), representing the sieve fraction with the highest mass concentration in the pre-milling feed. The characteristic product size L_2 was

taken as the sieve size at which the highest (or second-highest) post-milling mass was concentrated, as a simplified indicator of the dominant product fraction.

This approach represents a simplification: in a rigorous analysis, a weighted mean particle size (d_m) calculated from the full PSD would be used for both L_1 and L_2 . However, for the purposes of this investigation, the dominant-fraction method is used as a first-order approximation and is considered sufficient for comparative purposes.

The material constants used are: $K_r = 0.0179 \text{ m}^2/\text{J}$ and $F_c = 30 \text{ MPa}$ ($30 \times 10^6 \text{ Pa}$) for silica sand (Coulson & Richardson, Vol. 2). (K. M. A. S. Bandara, 2022) (Food Tech Notes, 2020)

Applying Equation 2.2:

$$E = K_r \cdot F_c \cdot (1/L_2 - 1/L_1) \quad (\text{Eq. 4.1})$$

Figure 3 energy calculation

Table 4.5 — Rittinger's Law Energy Calculations Summary

Grinding Medium	L_1 (μm)	L_2 (μm)	$1/L_2 - 1/L_1$ (m^{-1})	E (J/kg)
Steel Rods	1000	400	1500	805.5×10^6
Steel Balls	1000	600	667	358.0×10^6
Ceramic Balls	1000	600	667	358.0×10^6

The rod mill required approximately $805.5 \times 10^6 \text{ J/kg}$ of specific energy to achieve the observed size reduction, compared to approximately $358.0 \times 10^6 \text{ J/kg}$ for both ball mill configurations — a ratio of approximately 2.25:1. This result indicates that rod milling, while more energy-intensive in absolute terms, produces a meaningfully finer product ($L_2 = 400 \mu\text{m}$ versus $600 \mu\text{m}$), which represents a greater degree of size reduction from the same initial feed.

4.3 Discussion of Results

The particle size distribution data and energy calculations together support several conclusions regarding the comparative performance of the three grinding media.

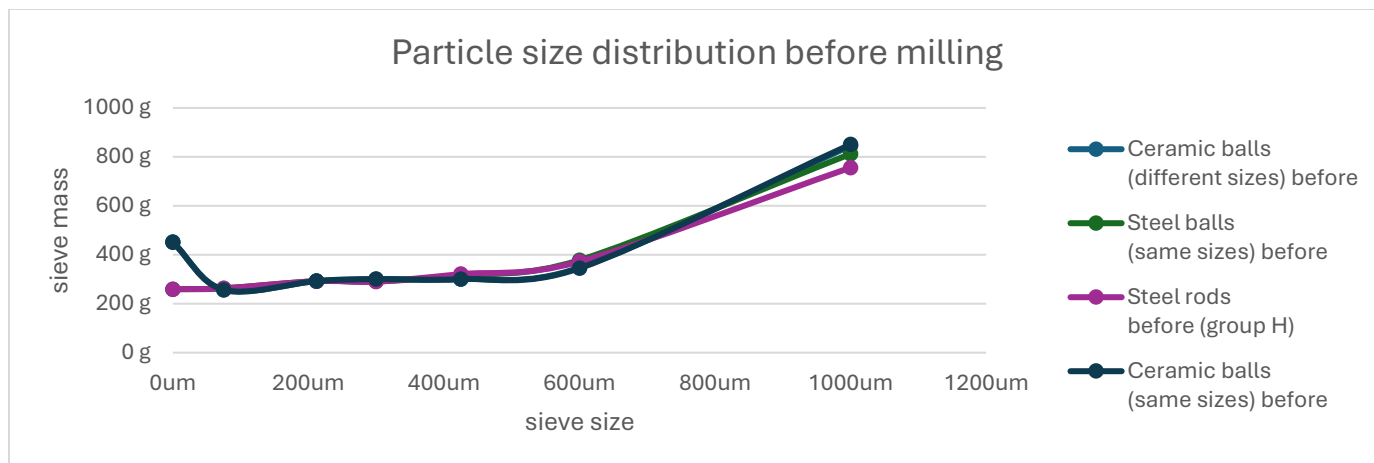


Figure 4 Particle size distribution before milling

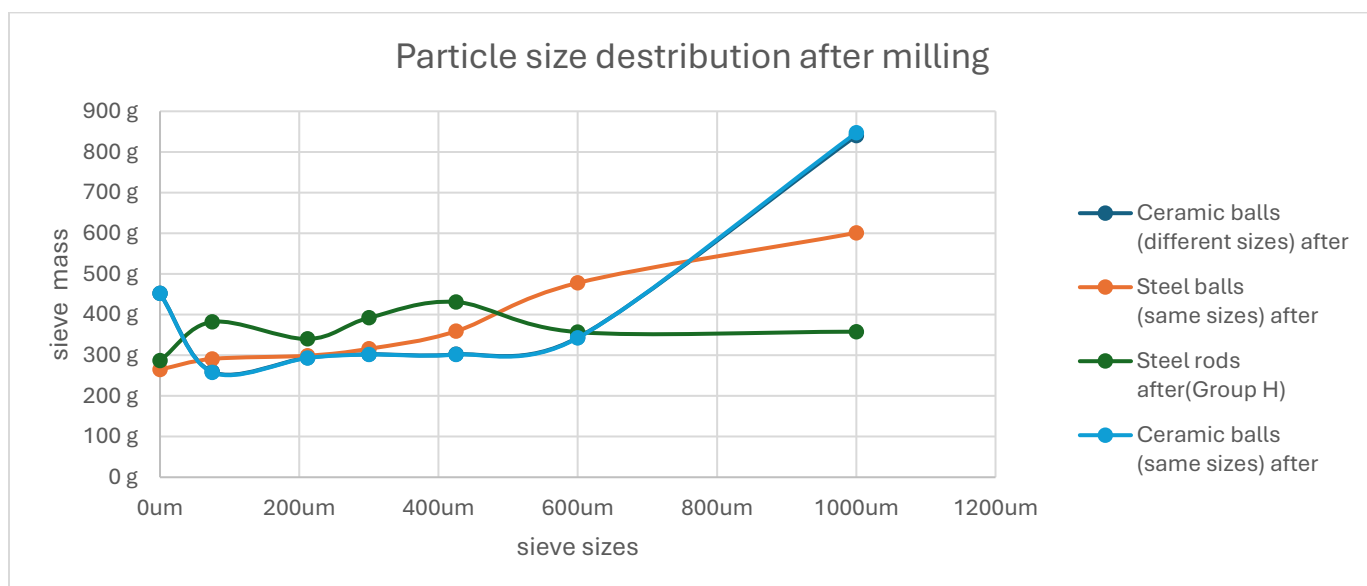


Figure 5 Particle size distribution after milling

4.3.1 Rod Mill Performance

The rod mill demonstrated the most effective size reduction of the three media tested, achieving a characteristic product size of approximately 400 μm from a feed of 1000 μm . Examination of the post-milling PSD reveals a notable concentration of mass in the 250–600 μm range, with relatively little material remaining at the coarse end of the distribution and limited production of very fine particles below 150 μm . This pattern is directly consistent with the screening effect described in

the literature: the parallel rod geometry selectively crushes larger particles while allowing fine particles to pass through without further grinding, resulting in a controlled product size band.

The limited movement in the ultra-fine fraction ($< 150 \mu\text{m}$) for the rod mill is also consistent with a known phenomenon in which very fine particles adhere to the rod surfaces, removing them from the active grinding zone and preventing further size reduction. This is not a deficiency of the rod mill per se, but rather a fundamental characteristic of its geometry at the lower end of its effective size range.

4.3.2 Steel Ball Mill Performance

The steel ball mill produced measurable size reduction, with the post-milling PSD showing a shift toward finer fractions relative to the feed. However, the characteristic product size of approximately $600 \mu\text{m}$ indicates less effective grinding than the rod mill under equivalent conditions (10 minutes, 500 g feed). The ball mill does not exhibit a screening mechanism; rather, size reduction proceeds through stochastic impact and attrition events, producing a broader size distribution with less preferential targeting of the coarse fraction.

4.3.3 Ceramic Ball Mill Performance

The ceramic ball mill demonstrated the least effective size reduction of the three media tested. The post-milling PSD showed minimal shift from the feed distribution, with the characteristic product size remaining at approximately $600 \mu\text{m}$ — comparable to the steel ball result despite the use of mixed-size balls, which theoretically should improve grinding coverage.

The primary explanation for the ceramic ball mill's underperformance is the difference in density and hardness between ceramic and steel. Steel has a density of approximately 7800 kg/m^3 , compared to approximately $3500\text{--}4000 \text{ kg/m}^3$ for typical alumina ceramic balls. Lower density directly reduces the kinetic energy imparted per collision ($E = \frac{1}{2}mv^2$), resulting in less effective breakage per impact event. Additionally, ceramic materials are generally less hard than steel under compressive loading, which may result in preferential wear of the grinding medium rather than fracture of the feed particles.

The use of mixed-size ceramic balls was intended to test whether a graded charge improves grinding efficiency by filling the interstitial spaces between larger balls. However, given the already limited energy transfer per collision, the size gradation effect was insufficient to compensate for

the material property disadvantage. For the feed material and size range tested, the density and hardness of the grinding medium appear to be the dominant determinants of grinding efficacy, with size gradation playing a secondary role.

4.4 Sources of Error and Limitations

Several sources of error were identified that may have influenced the experimental results:

- Material loss during transfer: Fine particles were visibly lost to the atmosphere upon opening the mill after grinding, particularly for the ball mill runs. This represents a systematic reduction in the measured post-milling mass and may have altered the apparent PSD by preferentially removing the finest fractions from the sample.
- Human error in sieve measurement: Mass measurements were performed manually using an electronic balance. Residual sand adhering to sieve surfaces or transfer containers may have introduced small systematic errors in the recorded masses.
- Cross-group data collection: Data for the rod mill and steel ball mill were obtained from concurrent experimental groups. While the same procedure was followed, minor differences in equipment calibration, operator technique, or environmental conditions between groups may have introduced inter-group variability.
- Simplified L_1/L_2 selection: The use of the dominant-fraction sieve size as the characteristic particle size, rather than a weighted mean diameter, introduces approximation error into the Rittinger's Law calculations. The comparative trends are unaffected, but the absolute energy values should be interpreted with caution.
- Fixed milling duration: The 10-minute milling time may not represent steady-state grinding conditions for all media. A kinetic study across multiple time intervals would provide a more complete picture of grinding rate and efficiency.

5. Conclusion and Recommendations

This investigation compared the grinding performance of steel rods, steel balls, and ceramic balls using silica sand as the feed material, employing particle size distribution analysis and Rittinger's Law as the primary analytical tools.

5.1 Conclusions

The following conclusions are drawn from the experimental results and energy calculations:

- Rod milling was the most effective grinding medium for silica sand in the 400–1000 μm range, producing a characteristic product size of approximately 400 μm and a concentrated product size distribution in the 250–600 μm band. This is consistent with the screening effect described in the literature.
- Steel ball milling produced moderate size reduction, achieving a characteristic product size of approximately 600 μm . The broader product size distribution reflects the non-selective impact mechanism of ball milling relative to the screening geometry of rod milling.
- Ceramic ball milling was the least effective medium tested, achieving no measurable improvement over the steel ball result despite the use of a graded (mixed-size) charge. The inferior performance is attributed to the lower density and hardness of ceramic relative to steel, which reduces the impact energy transferred per collision.
- Rittinger's Law calculations indicate that rod milling required approximately 2.25 times the specific energy input of the ball mill configurations. This higher energy consumption reflects the greater degree of size reduction achieved and is consistent with the relationship between surface area creation and energy input implied by the $n = 2$ exponent in Rittinger's formulation.
- Media density and hardness are identified as the primary determinants of grinding efficacy for the conditions tested, with media size gradation playing a secondary role.

5.2 Recommendations

Based on the findings of this investigation, the following recommendations are made:

- Rod mills should be preferred over ball mills when a uniform, controlled product size in the medium-fine range (250–600 μm) is the target, particularly for materials with a tendency toward agglomeration such as silica.

- Ceramic balls should not be used as a substitute for steel balls in applications where significant size reduction is required from a hard feed material, unless operating conditions are specifically designed to compensate for the lower media density.
- Future investigations should include a kinetic study of grinding efficiency over multiple time intervals to determine whether the observed performance differences are time-dependent or reflect fundamental capacity limits of each medium.
- A weighted mean particle diameter should be used in place of the dominant-fraction simplification for more accurate energy requirement estimates in follow-up work.
- Investigation of alternative feed materials with different hardness values (e.g., limestone vs. quartz) would allow the effect of feed hardness on the comparative performance of the three media to be quantified.

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